

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning

COURSE OUTCOME COVERED

CO4: *Evaluate model-based reinforcement learning approaches for prediction, planning, and policy optimization.(BL3,BL4,BL5)*

Assessment Tool Weightage: 40%

Dr G. A. SENTHIL

QUESTIONS: 25

Total Marks: 50

Annexure A

Design and Develop Model

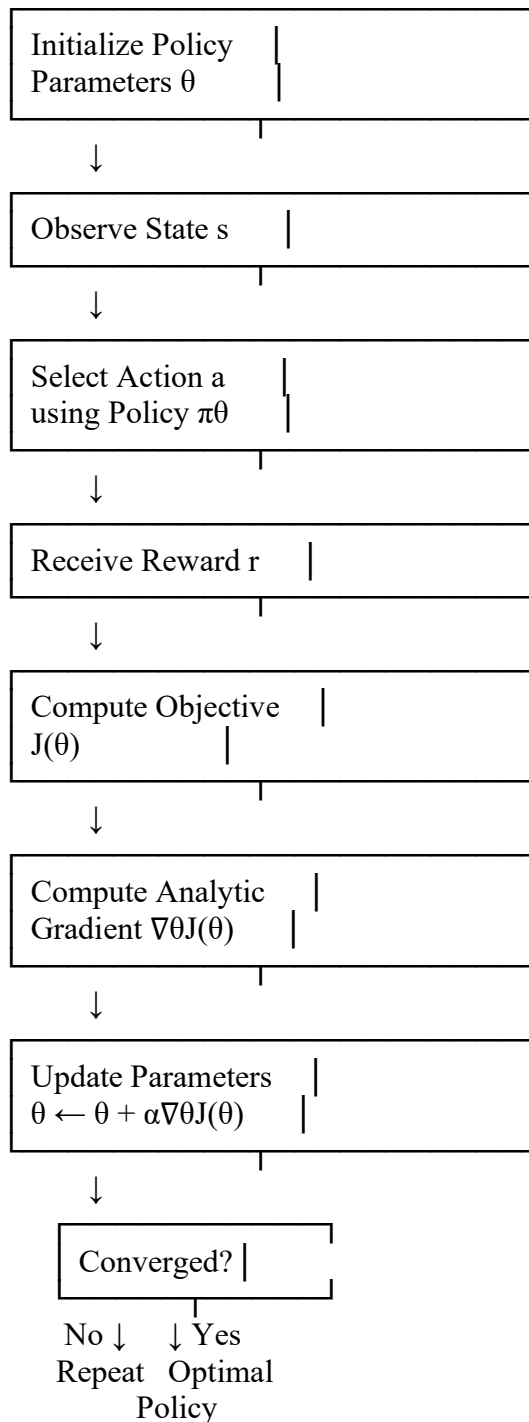
DURATION: 4 HRS (2 Sessions)

Analytic Gradient Computation

1. Draw a flowchart explaining the steps involved in analytic gradient computation for policy optimization.
2. Design a concept map showing how gradients are computed and propagated during policy learning.
3. Draw a block diagram illustrating gradient computation, parameter updates, and reward optimization in Reinforcement Learning.
4. Prepare a flowchart explaining gradient-based optimization using backpropagation in Reinforcement Learning.
5. Draw a concept map showing the relationship among policy parameters, gradients, optimization algorithms, and rewards.

ANSWERS:

1. Flowchart – Analytic Gradient Computation for Policy Optimization Diagram

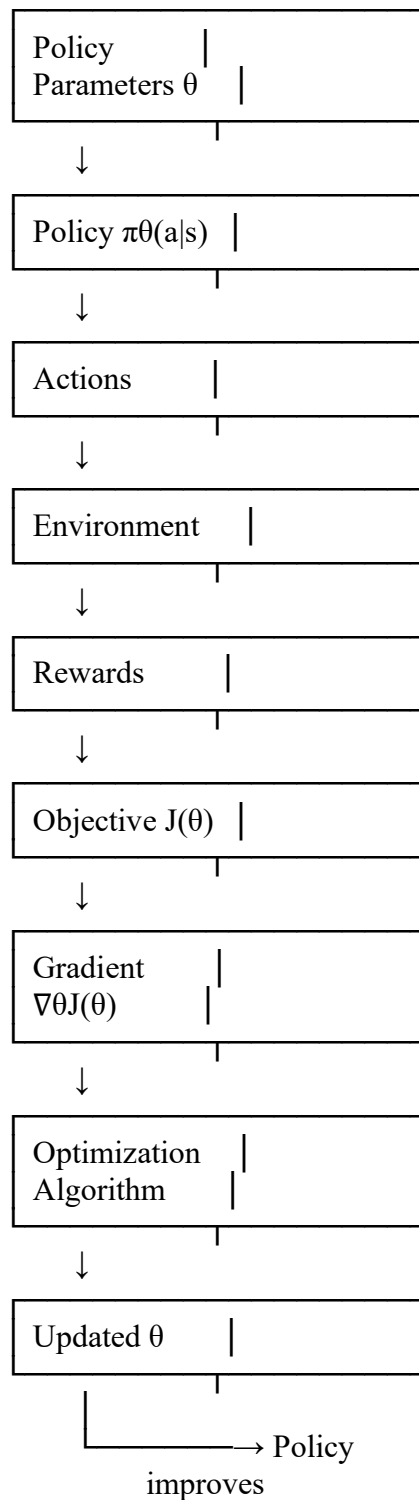


Explanation

Analytic gradient computation calculates the exact gradient of the policy objective with respect to policy parameters. The gradient is then used to update the policy toward higher expected rewards.

Key equation:

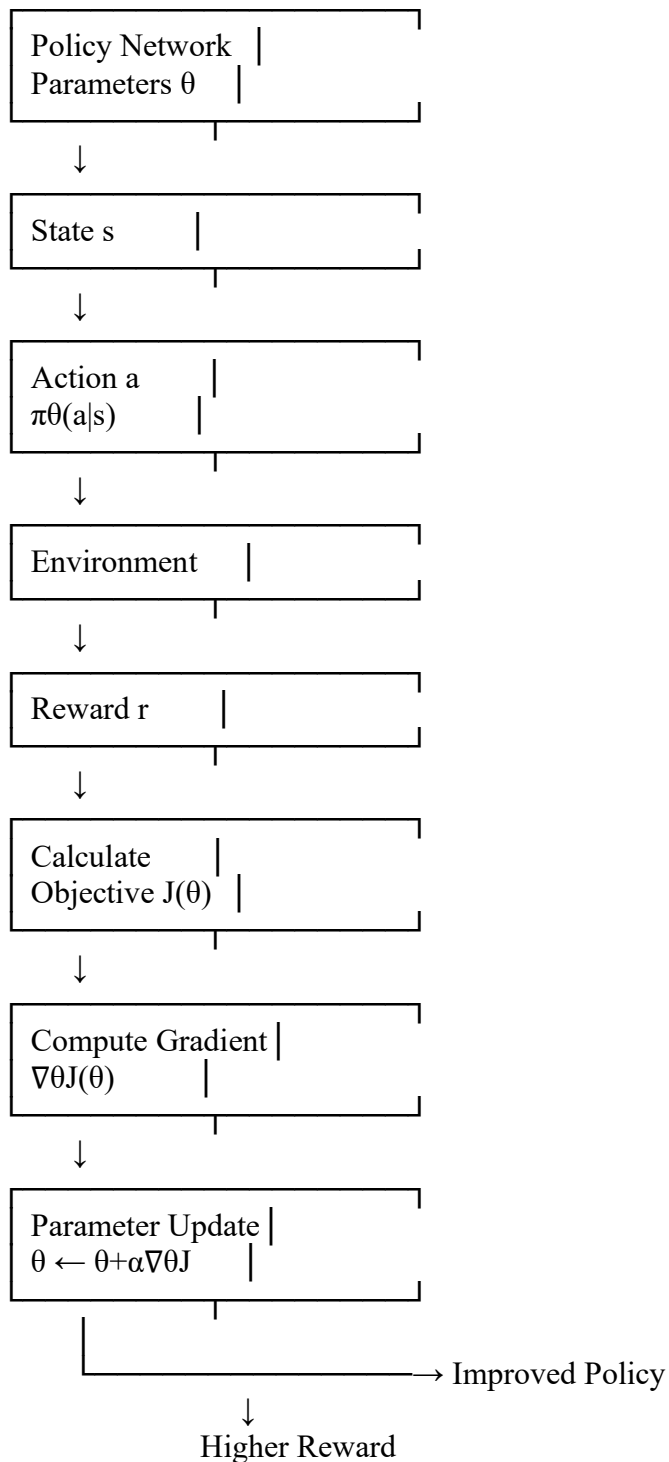
2. Concept Map – Gradients in Policy Learning Diagram



Main idea

Parameters $\rightarrow$ Policy $\rightarrow$ Actions $\rightarrow$ Rewards $\rightarrow$ Objective $\rightarrow$ Gradient $\rightarrow$ Optimization $\rightarrow$ Updated Parameters

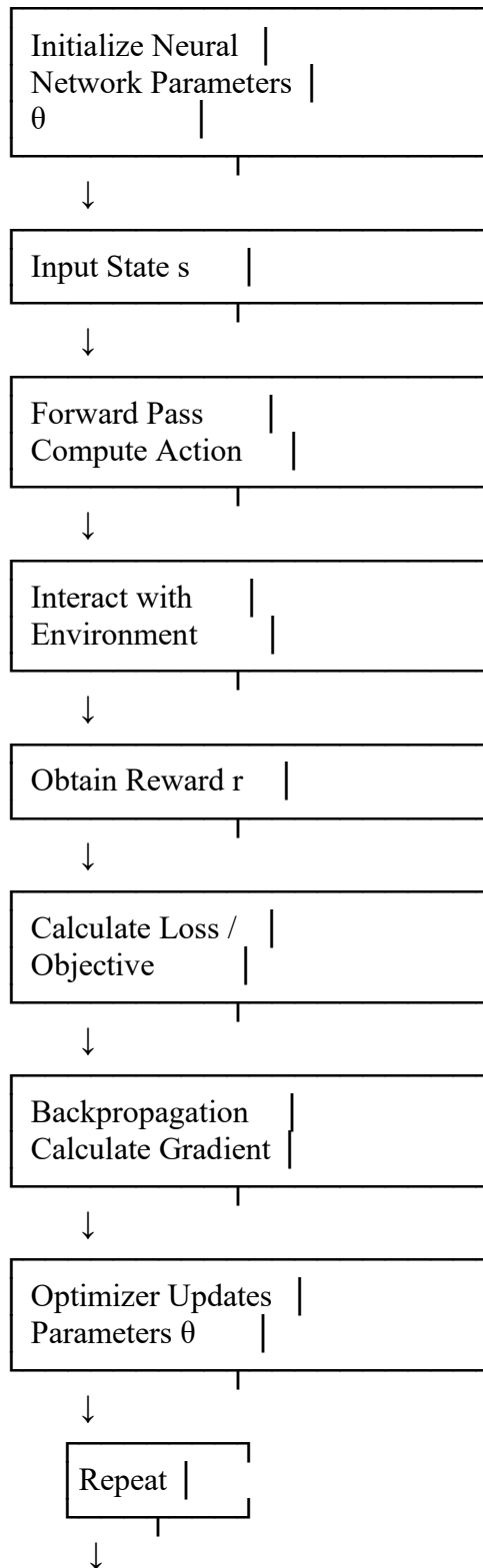
3. Block Diagram – Gradient Computation, Parameter Updates and Reward Optimization



Explanation

The agent uses its policy to select an action. The environment produces a reward. The gradient of the objective is calculated and used to update the policy parameters. Repeated updates aim to maximize the reward.

4. Flowchart – Gradient-Based Optimization Using Backpropagation in RL Diagram



Better Policy

Important equation

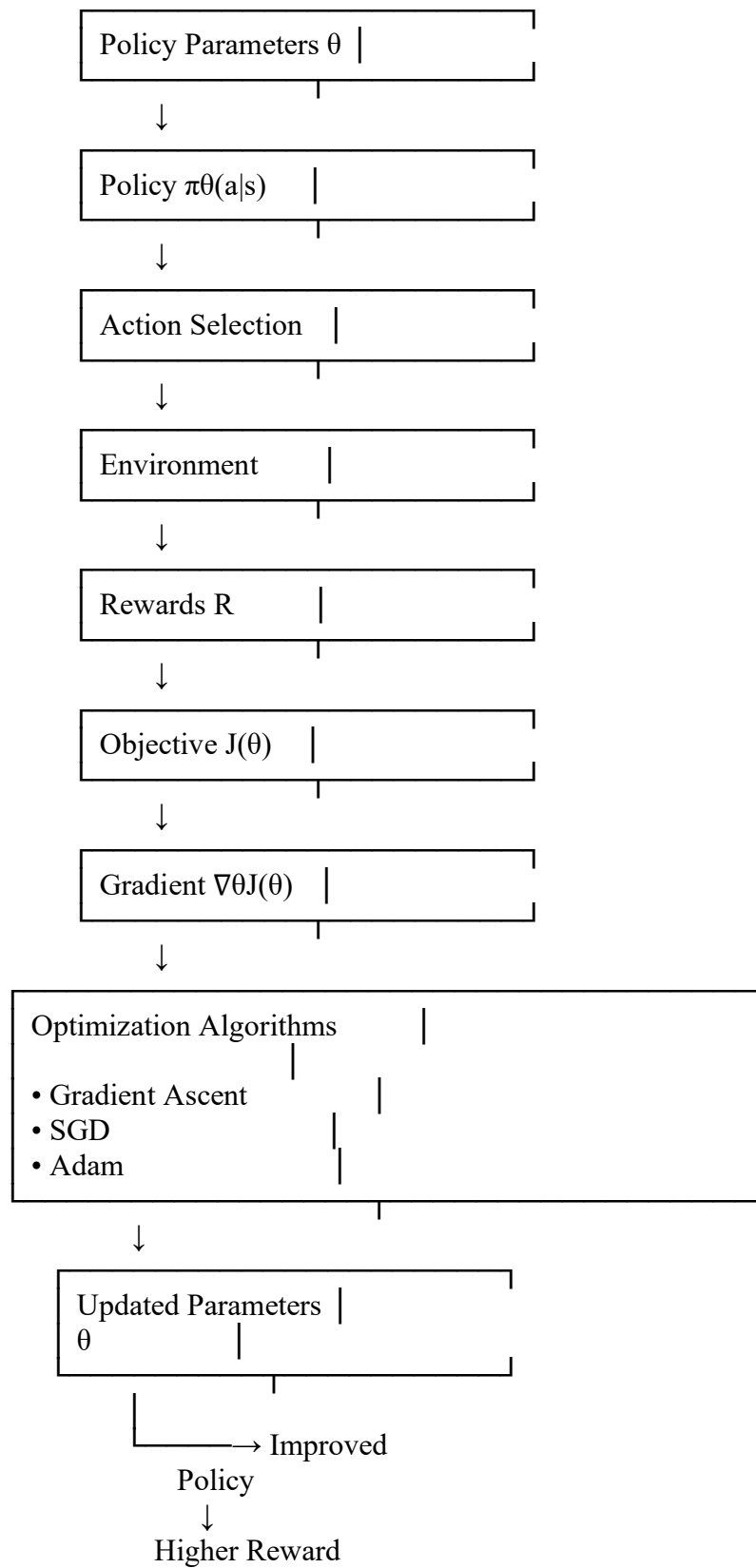
Using gradient descent:

For reward maximization, policy-gradient methods commonly use:

Main components

Forward Pass → Reward → Objective → Backpropagation → Gradient → Optimizer → Updated Policy

5. Concept Map – Policy Parameters, Gradients, Optimization Algorithms and Rewards



Code of Conduct:

*I **Kandukuri Kavya-192425283** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Kandukuri Kavya-192425283

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism ; uneven team participation	Lacks professionalism ; minimal contribution or ethical awareness